

Example Problem #1

A single phase power system for a small manufacturing facility is supplied by a 240V tie from the electric utility company. Inside the manufacturing facility, there are two power distribution panels to which all of the building's loads are connected. The first distribution panel is connected to the electric utility company tie through a 500ft power cable, and a 350ft power cable connects the two distribution panels together.

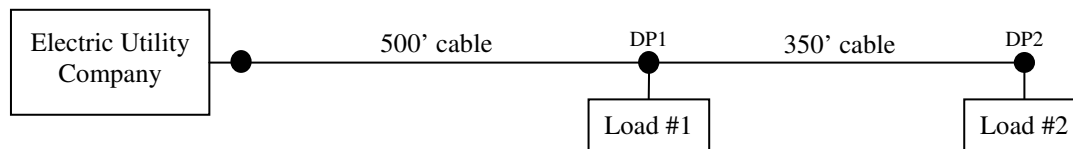
The net load connected to the first distribution panel has equivalent impedance of $300\angle 50^\circ \Omega$ and the net load connected to the second distribution panel has equivalent impedance of $20\angle 75^\circ \Omega$. The type of power cable used has 2-conductors with each conductor having distributed series impedance of $0.3\angle 20^\circ \Omega/1000\text{ft}$. The electric utility company tie has source impedance of $0.15\angle 85^\circ \Omega$.

Determine the following quantities for this system. Use the series impedance model for the power cables.

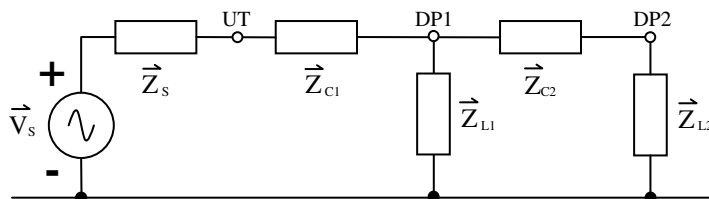
- Real and reactive power** supplied by the electric utility company to the manufacturing facility.
- Voltage** at each of the two distribution panels.
- Power system efficiency.**

Step 1: Visualize

Block Diagram



Step 2: Equivalent Circuit



Step 3: Circuit Parameters

$$\vec{V}_s = 240\angle 0^\circ \text{ V}, \quad \vec{Z}_s = 0.15\angle 85^\circ \Omega, \quad \vec{Z}_{L1} = 300\angle 50^\circ \Omega, \quad \vec{Z}_{L2} = 20\angle 75^\circ \Omega$$

$$\vec{Z}_C = 0.3\angle 20^\circ \Omega / \text{kft}, \quad \vec{Z}_{C1} = 2 \cdot \left(\frac{500 \text{ ft}}{1000 \text{ ft} / \text{kft}} \right) \vec{Z}_C = 0.3\angle 20^\circ \Omega, \quad \vec{Z}_{C2} = 2 \cdot \left(\frac{350 \text{ ft}}{1000 \text{ ft} / \text{kft}} \right) \vec{Z}_C = 0.21\angle 20^\circ \Omega$$

Step 4: Circuit Analysis

$$\vec{Z}_{eq1} = \left[\vec{Z}_{L1}^{-1} + (\vec{Z}_{C2} + \vec{Z}_{L2})^{-1} \right]^{-1} = 18.9574\angle 73.01^\circ \Omega, \quad \vec{I}_s = \vec{V}_s / (\vec{Z}_s + \vec{Z}_{C1} + \vec{Z}_{eq1}) = 12.4444\angle -72.39^\circ \text{ A}$$

$$\vec{V}_{UT} = \vec{V}_s - \vec{Z}_s \vec{I}_s = 238.1787\angle -0.098^\circ \text{ V}, \quad \vec{S}_{UT} = \vec{V}_{UT} \vec{I}_s^* = 2963.9995\angle 72.29^\circ \text{ VA}$$

$$\vec{V}_{DP1} = \vec{Z}_{eq1} \vec{I}_s = 235.9137\angle 0.619^\circ \text{ V}, \quad \vec{V}_{DP2} = \vec{V}_{DP1} \left(\frac{\vec{Z}_{L2}}{\vec{Z}_{C2} + \vec{Z}_{L2}} \right) = 234.4928\angle 1.109^\circ \text{ V}$$

$$\vec{S}_{L1} = V_{DP1}^2 / \vec{Z}_{L1}^* = 185.5176\angle 50^\circ \text{ VA}, \quad \vec{S}_{L2} = V_{DP2}^2 / \vec{Z}_{L2}^* = 2749.3443\angle 75^\circ \text{ VA}$$

Step 5: Final Results

- Real and reactive power** supplied by the electric utility company to the manufacturing facility.

$$P_{UT} = \text{Re}\{\vec{S}_{UT}\} = 901.62 \text{ W}, \quad Q_{UT} = \text{Im}\{\vec{S}_{UT}\} = 2823.54 \text{ VAR}$$

- Voltage** at each of the two distribution panels.

$$V_{DP1} = |\vec{V}_{DP1}| = 235.91 \text{ V}, \quad V_{DP2} = |\vec{V}_{DP2}| = 234.49 \text{ V}$$

- Power system efficiency.**

$$P_{L1} = \text{Re}\{\vec{S}_{L1}\} = 119.248 \text{ W}, \quad P_{L2} = \text{Re}\{\vec{S}_{L2}\} = 711.583 \text{ W}, \quad \eta = \left(\frac{P_{L1} + P_{L2}}{P_{UT}} \right) \times 100 = 92.15\%$$

Example Problem #2

A three phase power system for a small manufacturing facility is supplied by a 485V tie from the electric utility company. Inside the manufacturing facility, there are three power distribution panels to which all of the building's loads are connected. The first distribution panel is connected to the electric utility company tie through a 500ft power cable, and the second and third distribution panels are connected to the first distribution panel through separate 350ft and 475ft power cables.

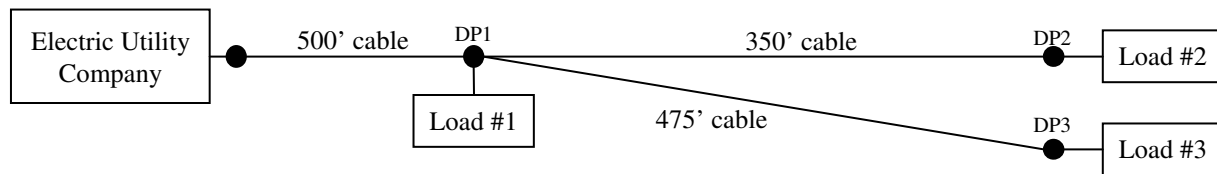
The net load connected to the first distribution panel has an equivalent Y-connected per phase impedance of $500\angle 40^\circ \Omega$, and the net loads connected to both the second and third distribution panels each have an equivalent Δ -connected impedance of $75\angle 70^\circ \Omega$. The type of power cable used has 4-conductors with each conductor having a distributed series impedance of $0.0003\angle 20^\circ \Omega/\text{ft}$. The electric utility company tie has a per phase source impedance of $0.09\angle 83^\circ \Omega$.

Determine the following quantities for this system. Use the series impedance model for the power cables.

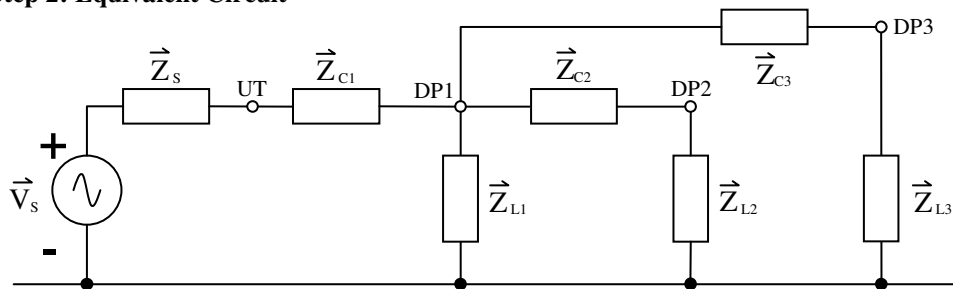
- Real and reactive power** supplied by the electric utility company to the manufacturing facility.
- Voltage** at each of the three distribution panels.

Step 1: Visualize

Block Diagram



Step 2: Equivalent Circuit



Step 3: Circuit Parameters

$$\vec{V}_s = \left(\frac{485}{\sqrt{3}} \right) \angle 0^\circ \text{ V}, \quad \vec{Z}_s = 0.09 \angle 83^\circ \Omega, \quad \vec{Z}_{L1} = 500 \angle 40^\circ \Omega, \quad \vec{Z}_{L2} = \vec{Z}_{L3} = \frac{1}{3} (75 \angle 70^\circ \Omega), \quad \vec{Z}_C = 0.0003 \angle 20^\circ \Omega / \text{ft}$$

$$\vec{Z}_{C1} = 500 \vec{Z}_C = 0.15 \angle 20^\circ \Omega, \quad \vec{Z}_{C2} = 350 \vec{Z}_C = 0.105 \angle 20^\circ \Omega, \quad \vec{Z}_{C3} = 475 \vec{Z}_C = 0.1425 \angle 20^\circ \Omega$$

Step 4: Circuit Analysis

$$\vec{Z}_{eq1} = \left[\vec{Z}_{L1}^{-1} + (\vec{Z}_{C2} + \vec{Z}_{L2})^{-1} + (\vec{Z}_{C3} + \vec{Z}_{L3})^{-1} \right]^{-1} = 12.27181 \angle 69.085^\circ \Omega$$

$$\vec{I}_s = \frac{\vec{V}_s}{\vec{Z}_s + \vec{Z}_{C1} + \vec{Z}_{eq1}} = 22.4772 \angle -68.663^\circ \text{ A}, \quad \vec{V}_{UT} = \vec{V}_s - \vec{Z}_s \vec{I}_s = 278.0554 \angle -0.103^\circ \text{ V}$$

$$\vec{S}_{UT} = \vec{V}_{UT} \vec{I}_s^* = 6249.8968 \angle 68.560^\circ \text{ VA}, \quad \vec{V}_{DP1} = \vec{Z}_{eq1} \vec{I}_s = 275.8355 \angle 0.422^\circ \text{ V}$$

$$\vec{V}_{DP2} = \vec{V}_{DP1} \left(\frac{\vec{Z}_{L2}}{\vec{Z}_{C2} + \vec{Z}_{L2}} \right) = 275.0915 \angle 0.606^\circ \text{ V}, \quad \vec{V}_{DP3} = \vec{V}_{DP1} \left(\frac{\vec{Z}_{L3}}{\vec{Z}_{C3} + \vec{Z}_{L3}} \right) = 274.8260 \angle 0.671^\circ \text{ V}$$

Step 5: Final Results

- Real and reactive power** supplied by the electric utility company to the manufacturing facility.

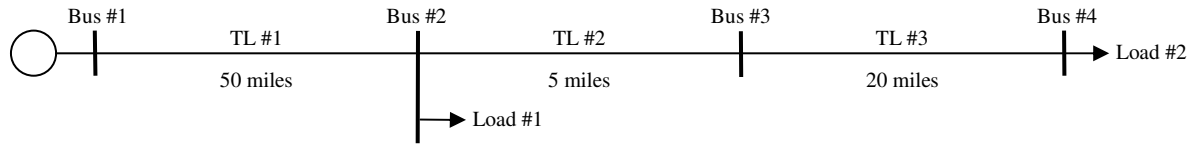
$$P_{3\phi UT} = 3 \cdot \text{Re} \{ \vec{S}_{UT} \} = 6853.54 \text{ W}, \quad Q_{3\phi UT} = 3 \cdot \text{Im} \{ \vec{S}_{UT} \} = 17,452.22 \text{ VAR}$$

- Voltage** at each of the three distribution panels.

$$V_{LLDP1} = \sqrt{3} \cdot |\vec{V}_{DP1}| = 477.76 \text{ V}, \quad V_{LLDP2} = \sqrt{3} \cdot |\vec{V}_{DP2}| = 476.47 \text{ V}, \quad V_{LLDP3} = \sqrt{3} \cdot |\vec{V}_{DP3}| = 476.01 \text{ V}$$

Example Problem #3

The one line diagram for a high voltage 3 ϕ power system is shown below. In this system, a 232KV source supplies power to two loads through a network of transmission lines. Load #1 is Δ -connected with per phase impedance of $12,300\angle 10^\circ \Omega$, and Load #2 is Y-connected with per phase impedance of $3950\angle 75^\circ \Omega$. The transmission lines in this system all have distributed per phase series impedance of $0.5372\angle 84^\circ \Omega/\text{mile}$.

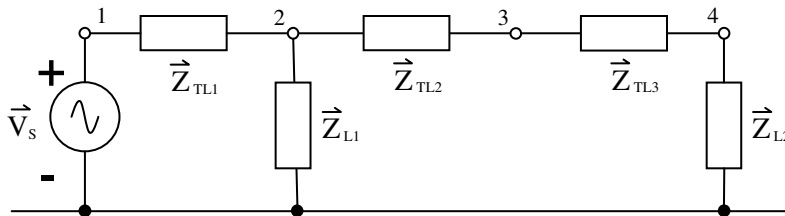


Determine the following quantities for this system. Use the series impedance model for the power cables.

- Real and reactive power** supplied by the source to the system.
- Voltage** at each of the four busses.

Step 1: Visualize (a one line diagram was given, so no other diagram needs to be drawn)

Step 2: Equivalent Circuit



Step 3: Circuit Parameters

$$\vec{V}_s = \left(\frac{232,000}{\sqrt{3}} \right) \angle 0^\circ V, \quad \vec{Z}_{L1} = \frac{1}{3}(12,300\angle 10^\circ \Omega), \quad \vec{Z}_{L2} = 3950\angle 75^\circ \Omega$$

$$\vec{Z}_{TL} = 0.5372\angle 84^\circ \Omega / \text{ft}, \quad \vec{Z}_{TL1} = 50\vec{Z}_{TL}, \quad \vec{Z}_{TL2} = 5\vec{Z}_{TL}, \quad \vec{Z}_{TL3} = 20\vec{Z}_{TL}$$

Step 4: Circuit Analysis

$$\vec{Z}_{eq3} = \vec{Z}_{TL3} + \vec{Z}_{L2} = 3960.6121\angle 75.024^\circ \Omega$$

$$\vec{Z}_{eq2} = \left[\vec{Z}_{L1}^{-1} + (\vec{Z}_{TL2} + \vec{Z}_{eq3})^{-1} \right]^{-1} = 2389.7080\angle 43.135^\circ \Omega$$

$$\vec{I}_s = \frac{\vec{V}_s}{\vec{Z}_{TL1} + \vec{Z}_{eq2}} = 55.5770\angle -43.552^\circ A, \quad \vec{S}_s = \vec{V}_s \vec{I}_s^* = 7,444,274.436\angle 43.552^\circ VA$$

$$\vec{V}_2 = \vec{Z}_{eq2} \vec{I}_s = 132,812.776\angle -0.418^\circ V$$

$$\vec{I}_{TL2} = \frac{\vec{V}_2}{\vec{Z}_{TL2} + \vec{Z}_{eq3}} = 33.5109\angle -75.448^\circ A$$

$$\vec{V}_3 = \vec{Z}_{eq3} \vec{I}_{TL2} = 132,723.8675\angle -0.424^\circ V, \quad \vec{V}_4 = \vec{Z}_{L2} \vec{I}_{TL2} = 132,368.2466\angle -0.448^\circ V$$

Step 5: Final Results

- Real and reactive power** supplied by the source to the system.

$$P_{3\phi S} = 3 \cdot \text{Re}\{\vec{S}_s\} = 16,185,617 W, \quad Q_{3\phi S} = 3 \cdot \text{Im}\{\vec{S}_s\} = 15,387,683 VAR$$

- Voltage** at each of the four busses.

$$V_{LL1} = \sqrt{3} \cdot |\vec{V}_s| = 232,000 V, \quad V_{LL2} = \sqrt{3} \cdot |\vec{V}_2| = 230,038 V$$

$$V_{LL3} = \sqrt{3} \cdot |\vec{V}_3| = 229,884 V, \quad V_{LL4} = \sqrt{3} \cdot |\vec{V}_4| = 229,269 V$$